Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 25: One Is Enough

Counterexamples

Chartrand & Zhang, Section 3.5, pp. 102–105

Name: _____ Date: _____

Where we are

Four weeks of this unit have been about proving things true. Today is about proving things *false* — which is usually far less work, provided you know exactly what a counterexample has to do. Most wrong answers here are not wrong arithmetic. They are counterexamples that do not actually counter anything.

By the end of today you should be able to

1. Say precisely what a counterexample to an implication must satisfy.
2. Disprove a false universal statement, including a biconditional.
3. Recognise a "counterexample" that proves nothing.

1. What a counterexample must do

A statement $\forall x \in S,\ P(x) \Rightarrow Q(x)$ is false exactly when its negation is true:

$$\sim\big(\forall x \in S,\ P(x) \Rightarrow Q(x)\big) \ \equiv\ \exists x \in S,\ P(x) \wedge (\sim Q(x))$$

That is the quantifier negation from Week 4 combined with Theorem 1.50 from Week 2.

Counterexample

A **counterexample** to $\forall x \in S,\ P(x) \Rightarrow Q(x)$ is an element $a \in S$ for which $P(a)$ is **true** and $Q(a)$ is **false**.

Both halves are required

An element where $P(a)$ is *false* is not a counterexample — it lands in a row where the implication is *true*, so it supports the claim rather than refuting it.
When you offer a counterexample, **check the hypothesis holds** before checking that the conclusion fails. That order catches most errors.

2. Worked examples

Example 3.30. Disprove: *let $n \in \mathbb{Z}$; if $4n+5$ is odd, then n is even.*

Try $n = 1$. Hypothesis: $4(1)+5 = 9$, which is odd — ✓. Conclusion: $n = 1$ is not even — fails. So $n = 1$ is a counterexample.

In fact every odd n works, since $4n + 5$ is always odd.

A fresh one. Disprove: *let $n \in \mathbb{Z}$; if $n^2 > 4$, then $n > 2$.*

Counterexample: $n = -3$. Then $n^2 = 9 > 4$, but $n > 2$ is false.

Where to look

Positive-integer intuition hides this one: every *positive* n with $n^2 > 4$ really does exceed 2. The counterexamples all live among the negatives.
Check the edges of the domain first: zero, negatives, fractions, the empty set. Those are where plausible claims break.

3. Disproving a biconditional

Example 3.31. Disprove: *let $m, n \in \mathbb{Z}$; then $3m + 7n$ is even if and only if m and n are both even.*

A biconditional is two implications. To disprove it, show at least one of them is false.

Implication	True or false?
If m, n both even, then $3m + 7n$ even	True.
If $3m + 7n$ even, then m, n both even	False: $m = n = 1$ gives $3 + 7 = 10$.

So $m = n = 1$ is a counterexample to the biconditional.

Say which direction fails

A good disproof names the failing direction. "$m = n = 1$ is a counterexample" is correct but thin; "the forward implication fails at $m = n = 1$, since $3m + 7n = 10$ is even while m and n are odd" is complete.

4. Counterexamples that prove nothing

Each attempted disproof below is wrong. Say why.

Claim and offered "counterexample"	What is wrong
"If n is divisible by 4, then n is even." Offered: $n = 3$.	$n = 3$ fails the hypothesis; the claim is actually true.
"If n^2 is odd, then $n > 0$." Offered: $n = 2$.	$n^2 = 4$ is even, so the hypothesis fails. Use $n = -1$.
"Every prime is odd." Offered: "most primes are odd."	Not an element at all. Name one: 2.

5. Proof or counterexample?

When a problem says *prove or disprove*, you must decide first — and the decision itself is part of the work.

A working method

1. Test several values, *including edge cases*.
2. If any fails with the hypothesis true, you have a counterexample. Stop.
3. If all pass, you have evidence, not proof — now try to prove it.

Try it. Prove or disprove: *let $n \in \mathbb{Z}$; if n is divisible by 4, then n is divisible by 8.*

False: $n = 4$ is divisible by 4 but not by 8.

Exit ticket

1. What two conditions must a counterexample to $P(x) \Rightarrow Q(x)$ satisfy?

2. Disprove: *if x is a real number, then $x^2 > x$.*

Before Wednesday

Read Section 3.6, "Existence Proofs," pp. 105–108.

Self-check (not collected). Section 3.5, Exercises 1, 3, 5, and 13 — odd-numbered, answers in the back.

Coming up

Homework 9 is due Friday, October 23, 11:59 p.m.
Quiz 8 is Monday, October 26 — Sections 3.4, 3.5, and 3.6.

Questions I still have